

Kotaro Ikeda

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About

My name is Kotaro Ikeda. I am interested in machine learning as well as its intersections with thermodynamics and dynamical systems.

Education

BE The University of Tokyo, Department of Mathematical Engineering and Information Physics, School of Engineering – Tokyo, Japan Apr 2022 – present

- Planned to graduate in March 2026
- Thesis advisor: Prof. Hiroshi Kori
- (unofficial) GPA: 3.87/4.3

Fundings

Funai Overseas Scholarship Sept 2026 – Aug 2028

Partial funding for studying abroad.

- \$36,000 per year for living cost
- \$14,000 per year for tuition
- full coverage of medical insurance

Publications

Pairwise Optimal Transports for Training All-to-All Flow-Based Condition Transfer Model Dec 2025

Kotaro Ikeda, Masanori Koyama, Jinzhe Zhang, Kohei Hayashi, Kenji Fukumizu
NeurIPS 2025 Poster (Acceptance rate: 24.52%)
[10.48550/arXiv.2504.03188](https://arxiv.org/abs/10.48550/arXiv.2504.03188)

Speed-accuracy relations for diffusion models: Wisdom from nonequilibrium thermodynamics and optimal transport June 2025

Kotaro Ikeda, Tomoya Uda, Daisuke Okanohara, Sosuke Ito
Physical Review X 15.3 (2025): 031031 (Top Journal in Physics [Impact Factor: 15.7])
[10.1103/x5vj-8jq9](https://arxiv.org/abs/10.1103/x5vj-8jq9)

Projects

Board Game agent development@May Fes Feb 2025 – May 2025

Created a reinforcement learning based agent for Blokus™

- Created a website and a real-world competition system for playing with our agents.
- Served as a project leader at [May Fes](#).

Research Experience Seminar on the Universal Principles of Life Apr 2023 – Mar 2024

Small-group seminar by Universal Biology Institute, the University of Tokyo

- Advised by Prof. Sosuke Ito
- Constructed a novel analogy between diffusion models and stochastic thermodynamics
- Theoretically discussed the optimality of noise schedules in diffusion models
- Published a paper on this topic in a top-physics journal
- Press Release is [here](#)

Experience

Part-time Engineer, Meltly, Inc. – Tokyo, Japan

Apr 2025 – present

- Directed training a generative model for anime-character images.
- Optimized multi-node distributed training (32x H100 GPUs) for fine-tuning Stable Diffusion 3.5, troubleshooting NCCL communication bottlenecks and memory efficiency.

Teaching Assistant, the University of Tokyo – Tokyo, Japan

Apr 2025 – Aug 2025

- Assisted in undergraduate course "Information"

Summer Internship, Preferred Networks, Inc. – Tokyo, Japan

Aug 2024 – Feb 2025

- Advised and mentored by Prof. Masanori Koyama and Prof. Kenji Fukumizu.
- Developed deep learning model for editing tasks utilizing flow matching and optimal transport theories.
- Achieved SOTA on QED optimization.
- Accepted at NeurIPS 2025

Part-time Engineer, Akari, Inc. – Tokyo, Japan

June 2023 – Mar 2024

- Enrolled in a project involving NeRF and 3D Gaussian splatting.

Skills

Programming: Proficient with Python, PyTorch and Git; good understanding of C++, Web development (React) and App development (Swift, Java)

Languages: English (fluent, TOEFL: 105/120), Japanese (native)

Conferences in Japan

Speed-accuracy trade-off for the diffusion models: Wisdom from nonequilibrium thermodynamics and optimal transport

Nov 2024

Kotaro Ikeda (presenter), Tomoya Uda, Daisuke Okanohara, Sosuke Ito

The 27th Information-Based Induction Sciences Workshop (IBIS), Saitama, Japan

GeoBeta Flow Matching

Nov 2024

Kotaro Ikeda (presenter), Masanori Koyama, Jinzhe Zhang, Kohei Hayashi, Kenji Fukumizu

The 27th Information-Based Induction Sciences Workshop (IBIS), Saitama, Japan